EXACTNESS IN LENGTH ONE FOR A MATRIX PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Complex attached to a matrix and its exactness in length one. defined in the same section, source lines 3780-3823, before the counted statement. Supplies the object of the statement.
- A matrix with an invertible maximal minor can be diagonalised by invertible row and column operations. used at proof lines 3843-3875. Is the reduction that makes the verification straightforward.
- Localisation preserves exactness of a complex of modules. used at proof lines 3875-3896 for the general case. Reduces the statement to the invertible-minor case.

Source reference. Original lemma 00MX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 839–839. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00MV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 837–837. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00MT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 835–835. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7067–7067. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05GP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7068–7068. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0548: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6631–6631. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0588: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6771–6771. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07DQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9420–9420. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00N1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 843–843. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a local ring. Suppose that $\varphi : R^m \to R^n$ is a map of finite free modules. The following are equivalent*

(1) *φ is injective,*
(2) *the rank of φ is m and the annihilator of $I(\varphi)$ in R is zero.*

If R is Noetherian these are also equivalent to

(3) *the rank of φ is m and either $I(\varphi) = R$ or it contains a nonzerodivisor.*

Here the rank of φ and $I(\varphi)$ are defined as in Algebra, Definition 00MV.

Proof. If any matrix coefficient of φ is not in $\mathfrak{m}$, then we apply Algebra, Lemma 00MT to write φ as the sum of $1 : R \to R$ and a map $\varphi' : R^{m-1} \to R^{n-1}$. It is easy to see that the lemma for φ' implies the lemma for φ. Thus we may assume from the outset that all the matrix coefficients of φ are in $\mathfrak{m}$.

Suppose φ is injective. We may assume $m > 0$. Let $\mathfrak{q} \in \text{WeakAss}(R)$ so that $R_\mathfrak{q}$ is an auto-associated ring. Then φ induces an injective map $R_\mathfrak{q}^m \to R_\mathfrak{q}^n$ which is universally injective by Lemmas 05GN and 05GP. Thus $\varphi : \kappa(\mathfrak{q})^m \to \kappa(\mathfrak{q})^n$ is injective. Hence the rank of φ mod $\mathfrak{q}$ is m and $I(\varphi \otimes \kappa(\mathfrak{q}))$ is not the zero ideal. Since m is the maximum rank φ can have, we conclude that φ has rank m as well (ranks of matrices can only drop after base change). Hence $I(\varphi) \cdot \kappa(\mathfrak{q}) = I(\varphi \otimes \kappa(\mathfrak{q}))$ is not zero. Thus $I(\varphi)$ is not contained in $\mathfrak{q}$. Thus none of the weakly associated primes of R are weakly associated primes of the R-module $\text{Ann}_R I(\varphi)$. Thus $\text{Ann}_R I(\varphi)$ has no weakly associated primes, see Algebra, Lemma 0548. It follows from Algebra, Lemma 0588 that $\text{Ann}_R I(\varphi)$ is zero.

Conversely, assume (2). The rank being m implies $n \geq m$. Write $I(\varphi) = (f_1, \ldots, f_r)$ which is possible as $I(\varphi)$ is finitely generated. By Algebra, Lemma 07DQ we can

find maps $\psi_i : R^n \to R^m$ such that $\psi \circ \varphi = f_i \mathrm{id}_{R^m}$. Thus $\varphi(x) = 0$ implies $f_i x = 0$ for $i = 1, \ldots, r$. This implies $x = 0$ and hence φ is injective.

For the equivalence of (1) and (3) in the Noetherian local case we refer to Algebra, Proposition 00N1. If the ring R is Noetherian but not local, then the reader can deduce it from the local case; details omitted. Another option is to redo the argument above using associated primes, using that there are finitely many of these, using prime avoidance, and using the characterization of nonzerodivisors as elements of a Noetherian ring not contained in any associated prime. □